Current Electricity

Current Electricity - Preparation Strategy For NEET

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good college all you need is rank. Many people fail to get a good rank because they lose marks in Physics. Out of 180 questions, 45 questions are from Physics. Each question is of 4 marks. Therefore, physics contains a weightage of 180 marks. For selection in NEET at least 510 marks are needed and for this, at least 90 to 110 marks should be scored from Physics. To score well you need good command in Electrostatics, Mechanics, Magnetism, Current Electricity, Magnetism, Simple Harmonic Motion, and Rotational Motion. Electricity alone covers almost 8-10% of the questions. These questions are basically conceptual questions and require speed in mathematical calculations. Current electricity is a part of NEET's Physics syllabus and it holds its importance in further studies too.

Basic Concepts Covered in Current Electricity

Current Electricity is one of the important chapters which seems easy to understand in the beginning but gets difficult once we proceed. It is a concept based chapter and needs a thorough understanding of the laws and requires ample practice. The chapter starts with the concept of current electricity and potential difference. Current electricity is basically the movement of free electrons in the conductor per unit time. This movement of electrons takes place only when a potential difference is applied at the two ends of the conductor. Therefore, the current is equal to charge (q) by time (t). Later, Ohm studied the relation between current and potential difference. Ohm's law states that "In a conductor, potential difference applied across the conductor is directly proportional to the amount of current flows." To remove this proportionality the concept of resistance was introduced as a constant. Resistance, in simple words, are the obstacle faced by the moving electrons in the conductor. This hindrance is offered by the conductor which obstructs the motion of electrons and hence reduces the current. Resistance is a constant which depends on the material of the conductor and increases with the increase in temperature or length of the conductor. Similarly, the flow of current can be increased by increasing the width of the conductor or wire which in turn reduces the resistance and gives enough passage to the flow of electrons. Considering these factors we can say that Resistance is directly proportional to the length of the wire or conductor and inversely proportional to the area of its cross-section. Again, to remove this proportionality the concept of resistivity was introduced as constant. Resistivity is also called specific resistance. Conductivity is the reciprocal (inverse) of resistivity.

The other important concepts are Drift Velocity and Current Density. Current density is the amount of current flowing per unit area of the cross-section around that particular point inside a conductor such that the area of the cross-section is normal to the direction of current. As the electrons flow opposite to the current, the movement of electrons gets accelerated which results in a collision. Therefore, the electrons acquire a small velocity which is called drift velocity. Drift velocity by the electric field is also called mobility. The other important concepts in this chapter are the combination of resistors and cells in a circuit and the process of finding the equivalent resistance and unknown values. Electromotive force, electric cell, internal resistance, emf of the cell, Kirchoff's law and its applications are important topics to be covered from the chapter current electricity. Apart from that, a lot of questions come from topics like Peltier Effect, Thomson Effect, Thermoelectric effect and Joule's Law Of Heating. Numericals and circuits diagrams mostly include the principles of Wheatstone Bridge, Meter Bridge and Potentiometer.

Last but not least, the idea of measuring instruments such as Galvanometer, Ammeter and Voltmeter turns out to be important not only in practical labs but also in competitive exams.

Right Approach to Current Electricity While Preparing for NEET

NEET is a high-level exam which comes after board exams. Therefore, the basics needed to be strengthened before going into the advanced level. It is not advisable to read theories in detail, instead, revise from notes and understand the concept. One of the other techniques is to club chapters in order to get the flow of the entire section at one go. For electricity, we can say that it is divided into two major sections - Static electricity and Current electricity. Static electricity is also known as electrostatics contains topics like the electric field, coulomb's law, potential, electric flux, gauss's law and capacitors whereas Current electricity consists of topics like ohm's law, resistance, resistivity, conductivity, resistor combinations, etc. In order to attain command over numerical questions, it is necessary to master Kirchoff's rules, resistance and cell combination rules by solving maximum numerical from books, sample papers and previous year's papers. Learn all important formulas and keep a note of important points. You can also make flashcards or sticky notes to help you in last-minute revision. Practise formulas and derivations to increase your speed because some of the NEET questions require a numerical answer for which calculations need to be done for a confirmed answer. Due to lack of time and speed students often give up on the answer or guess the answer if it is in the MCQ section. Therefore, a good speed is required to solve the problem in a limited period of time and get the right answer.

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